

Hospital Capacity Under Differ Conditions

A Monte Carlo SEIR Simulation — Critique & Extension of Sanvaliya et al. (2020)

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01

Background

The Original Study: Sanvaliya et al. (2020)

What they did

- ▶ SEIR compartmental model for COVID-19 in the Chicago area
- ▶ Monte Carlo simulation to project hospital bed demand over 60 days
- ▶ Two hypotheses tested: bed doubling & 25% lockdown
- ▶ Both hypotheses were reported as confirmed

This project

Type I: Critique and improvement of a published analysis

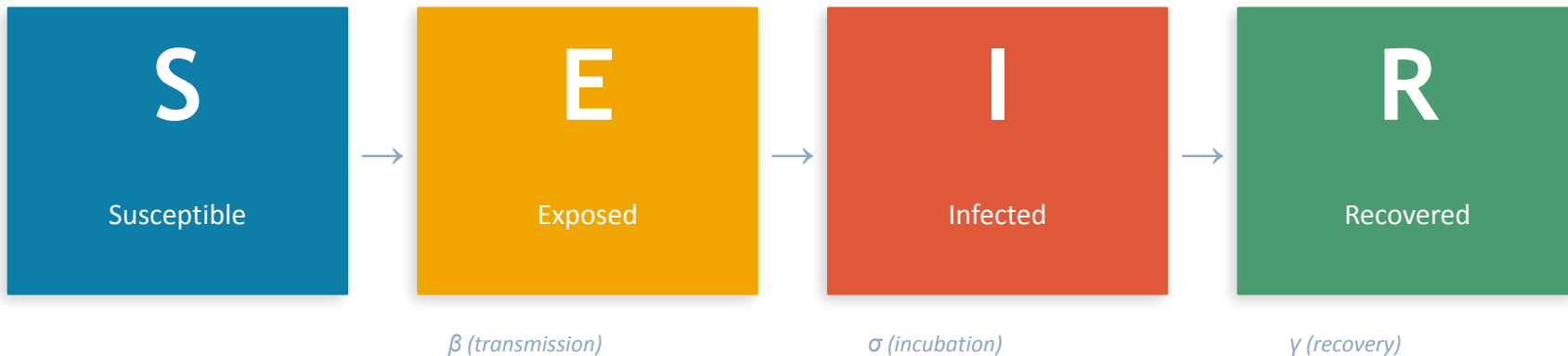
Scope: Chicago metro — 2.71M population, 33,000 hospital beds

Goal: Find what's broken, fix it, and ask harder questions

Research Question

"Does doubling hospital beds really eliminate overflow — or did the original model just have a bug?"

How the SEIR Model Works



From model → hospital beds

Each day, hospitalisations = $I \times \text{hospitalisation rate}$

Rate varies by variant: Original 3.5% | Delta 17% | Omicron 1% | Influenza 1.5%

Beds fill up → $\text{bed_occupancy} \geq \text{total_beds}$ → OVERFLOW

02

Revisions

Four Critical Defects in the Original Code

1 Integer casting bug

The transmission rate β (a decimal like 0.38) was cast to `int()`, producing zero. The susceptible compartment never decreased — so the model silently simulated no epidemic at all.

Impact: Core SEIR dynamics completely frozen

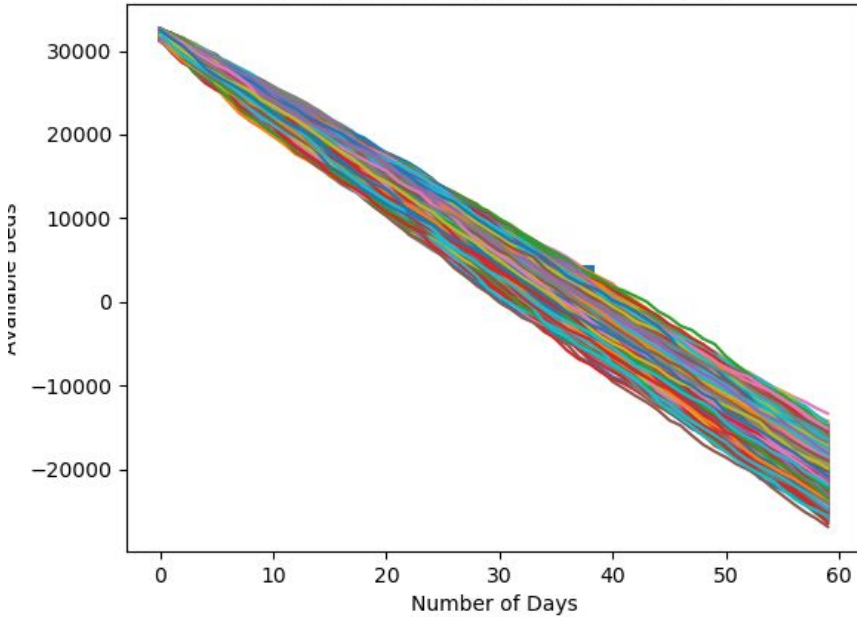
2 Population conservation violated

Exposed (E) was re-drawn as an independent random variable mid-loop, meaning $S + E + I + R \neq N$. People were being created and destroyed at every timestep.

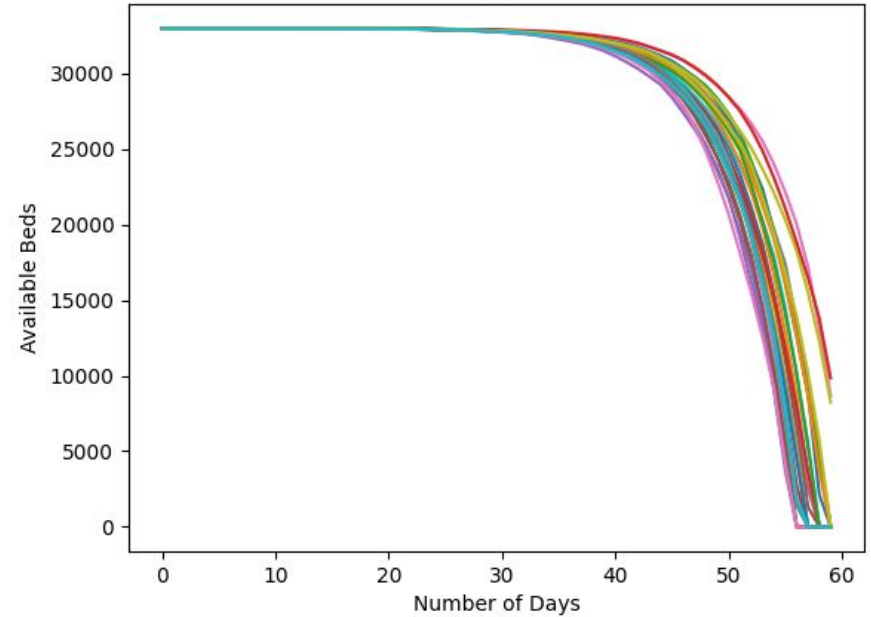
Impact: Simulation outcomes were physically meaningless

Why the Bugs Matter: The Numbers Change Dramatically

H1 Before: Available Number of Beds



Available Number of Beds



Why the Bugs Matter: The Numbers Change Dramatically

	Original Paper (buggy)	This Study (corrected)
H1: Overflow at baseline (33k beds)	Artificially low / near 0%	83.8% of runs overflow
H1: Overflow after doubling (66k beds)	"Eliminated" (0%)	Still 6.0% of runs overflow
H2: Effect of 25% lockdown	No discernible effect	Delta overflow drops 79.6% → 0%
H2: Effect on bed vacancy	~50% as claimed	~83.6% vacancy (more sensitive)

Bottom line: the bug made the original simulation look like a mild case. The corrected model behaves like an actual epidemic.

03

Advancements

Extended Model: New Research Questions

H1 Bed doubling under corrected SEIR

With the bug fixed, does doubling beds still "eliminate" overflow under a real Delta-class surge? The original claimed yes — we test this rigorously.

H2 Flu co-circulation with doubled beds

Even at 66,000 beds, adding circulating seasonal influenza on top of a COVID surge — does that erode the safety margin?

H3 Lockdown × variant × pathogen

The capacity of the hospital system is influenced by the extent of lockdowns, virus variants, and the flu.

Variant & Pathogen Parameters

Pathogen / Variant	R ₀ Range	Hosp. Rate	Initial Seed	Source
Original SARS-CoV-2	2 – 3	3.5%	1 case	WHO / CDC early estimates
Delta (B.1.617.2)	5 – 8	17%	1 case	Pearson et al. (2021)
Omicron (BA.1/BA.2)	10 – 16	1%	1 case	WHO TAG-VE (Jan 2022)
Influenza (H1N1/H3N2)	1.2 – 1.4	1.5%	1,000 cases	CDC FluView (seasonal)

Key distinction: Influenza is seeded at 1,000 active cases (endemic baseline), while COVID variants start with a single index case — matching the original early-pandemic framing.

Lockdown is modelled as: $\beta_{\text{effective}} = \beta \times (1 - L)$, where $L \in \{0, 0.25, 0.50\}$

04

Results

H1: Bed Doubling Under a COVID-Only Surge

Delta variant (R_0 5–8, hosp. rate 17%), no lockdown

83.8%

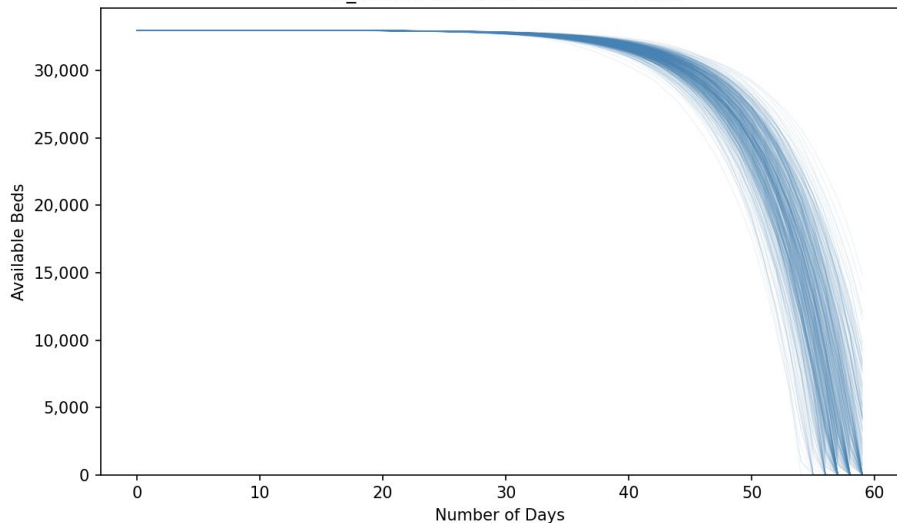


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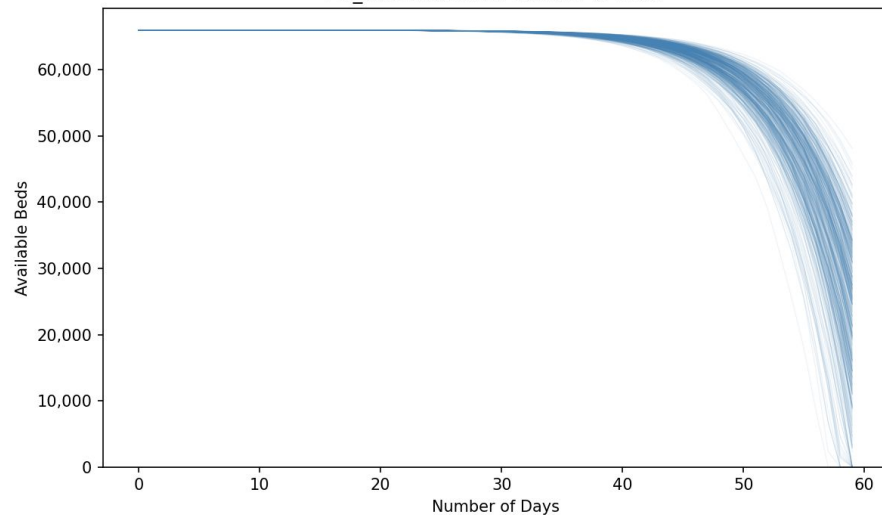
Overflow at 33k beds

Overflow at 66k beds

H1_before: Available Number of Beds



H1_after: Available Number of Beds



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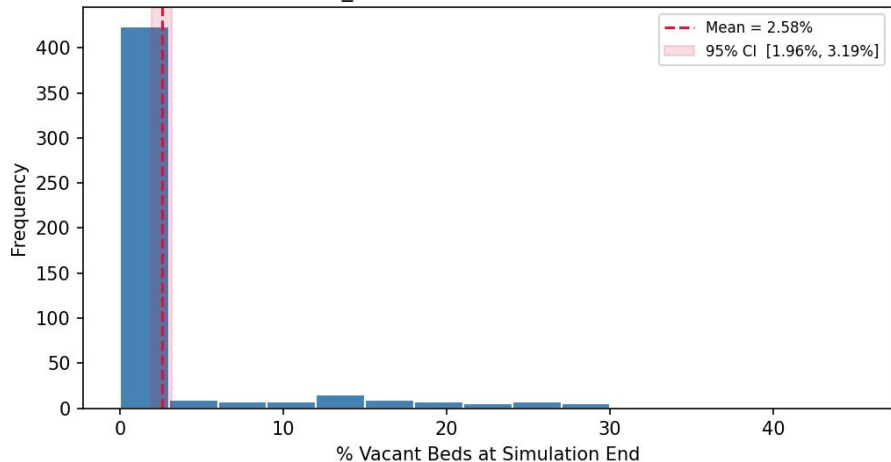


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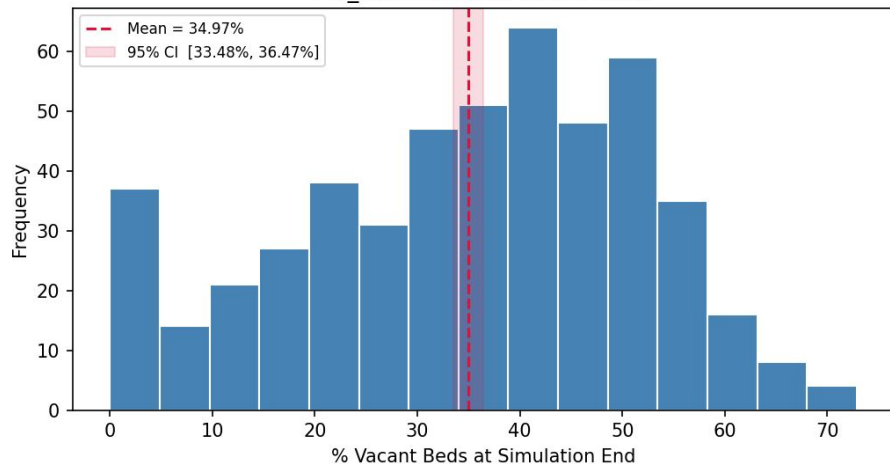
Overflow at 33k beds

Overflow at 66k beds

H1_before: Percent Vacant Beds



H1_after: Percent Vacant Beds



H2E: Does Flu Co-Circulation Erode the Safety Margin?

Both scenarios at 66,000 beds, Delta variant, no lockdown

COVID-only

Overflow probability

5.8%

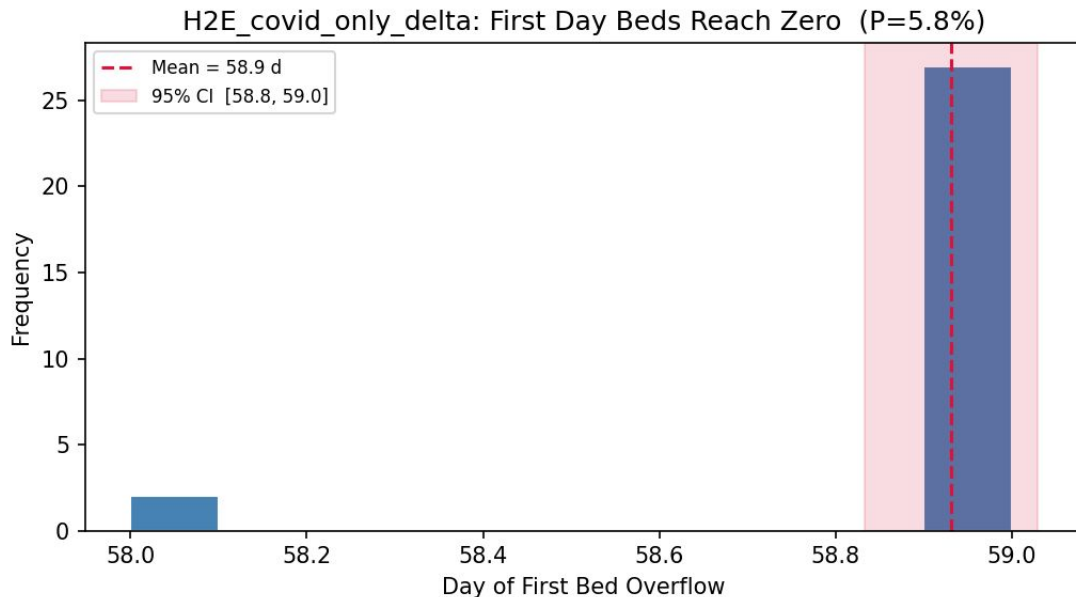
Mean vacancy: 34.4%

COVID + Flu

Overflow probability

7.2%

Mean vacancy: 29.9%



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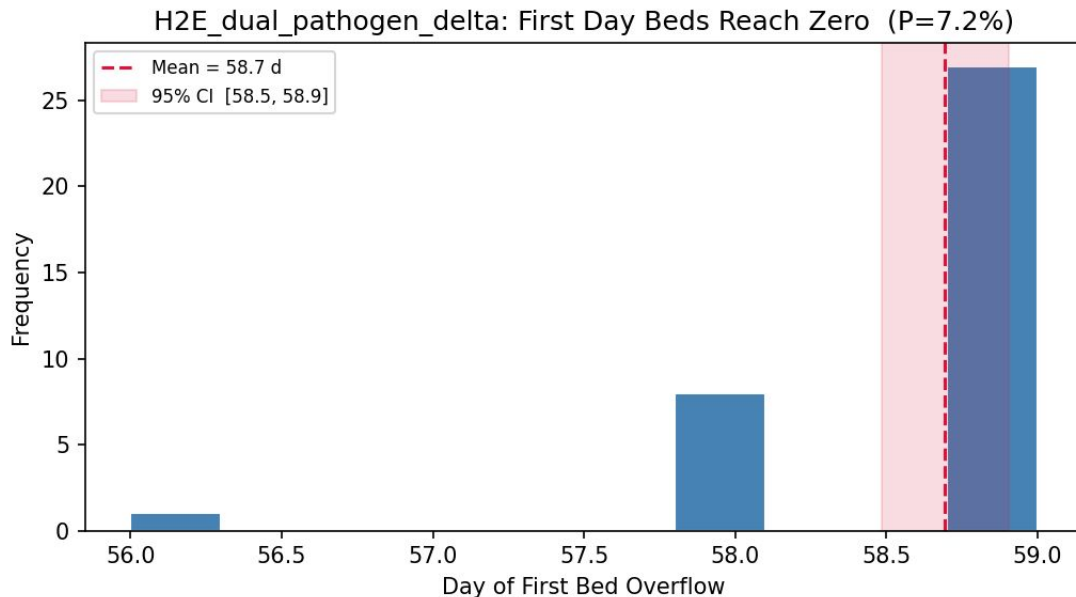
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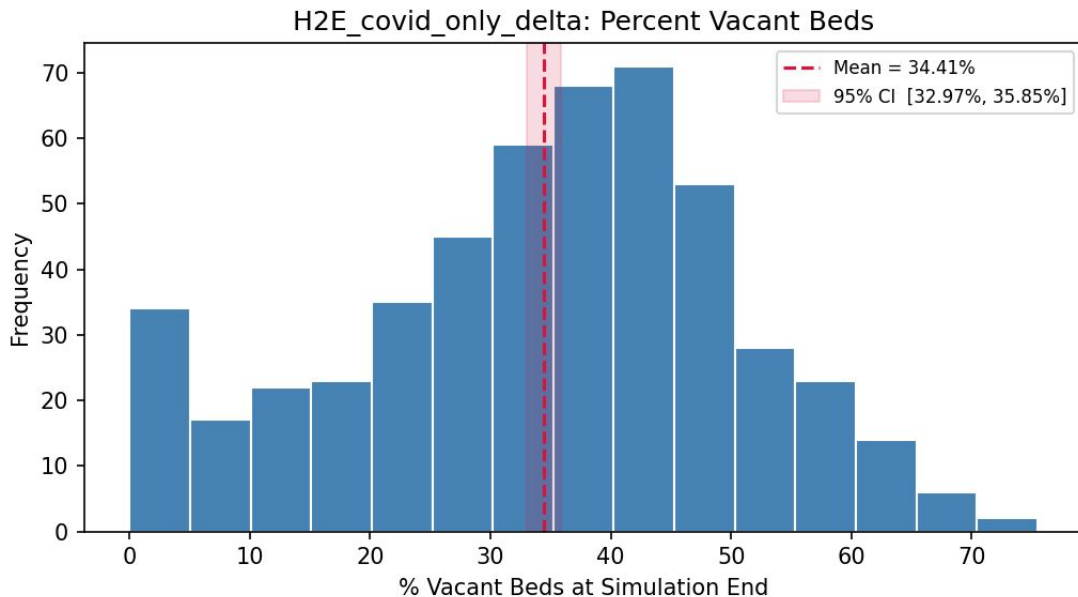
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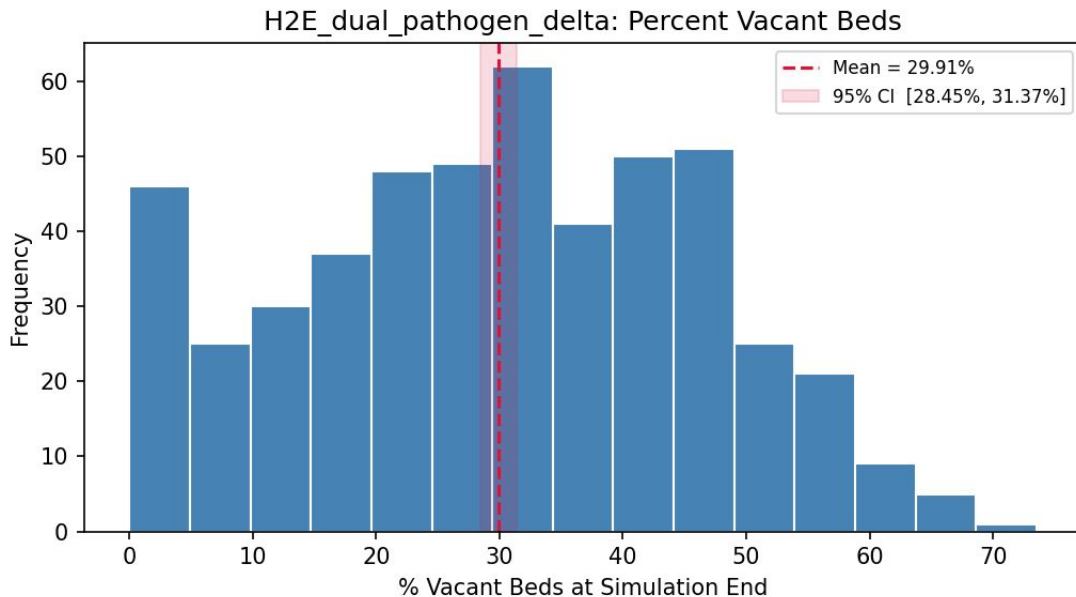
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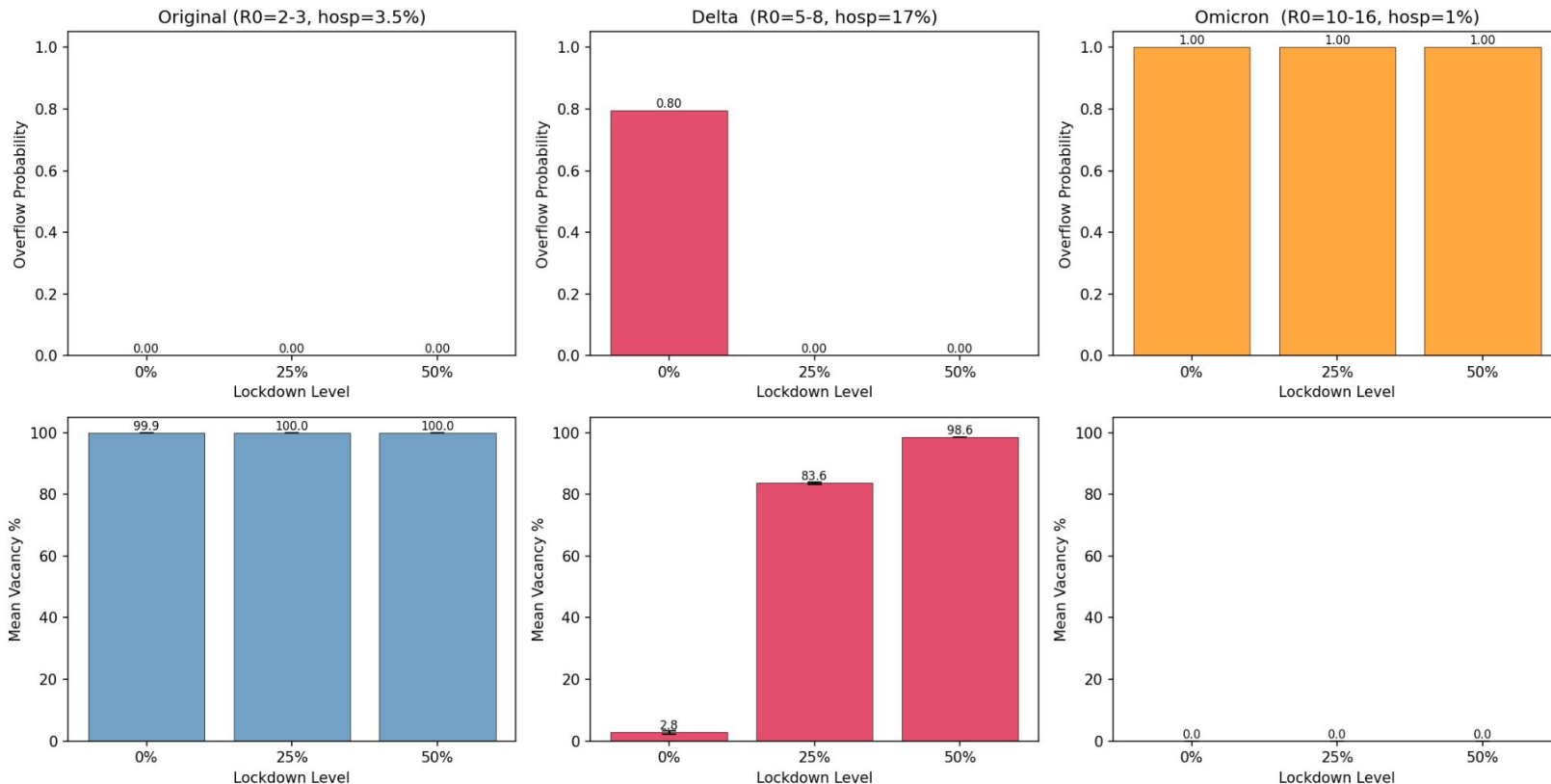


H3: Lockdown Effectiveness Depends Entirely on the Variant

	Lockdown 0%	Lockdown 25%	Lockdown 50%
Original	0.0% overflow	0.0% overflow	0.0% overflow
Delta	79.6% overflow	0.0% overflow	0.0% overflow
Omicron	100% overflow	100% overflow	100% overflow

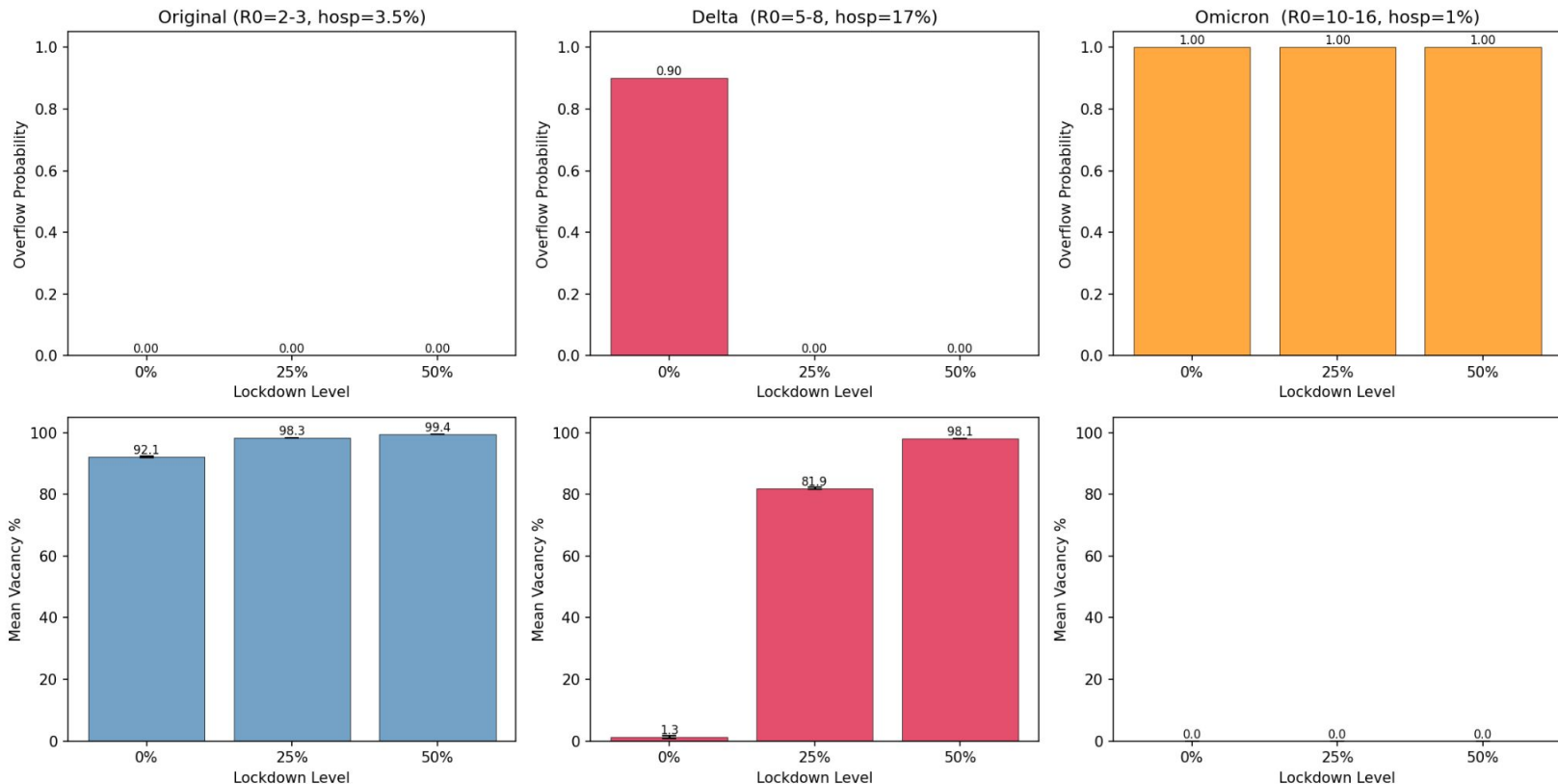
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COVID-only -- Lockdown Impact (3 Variants x Lockdown Levels)



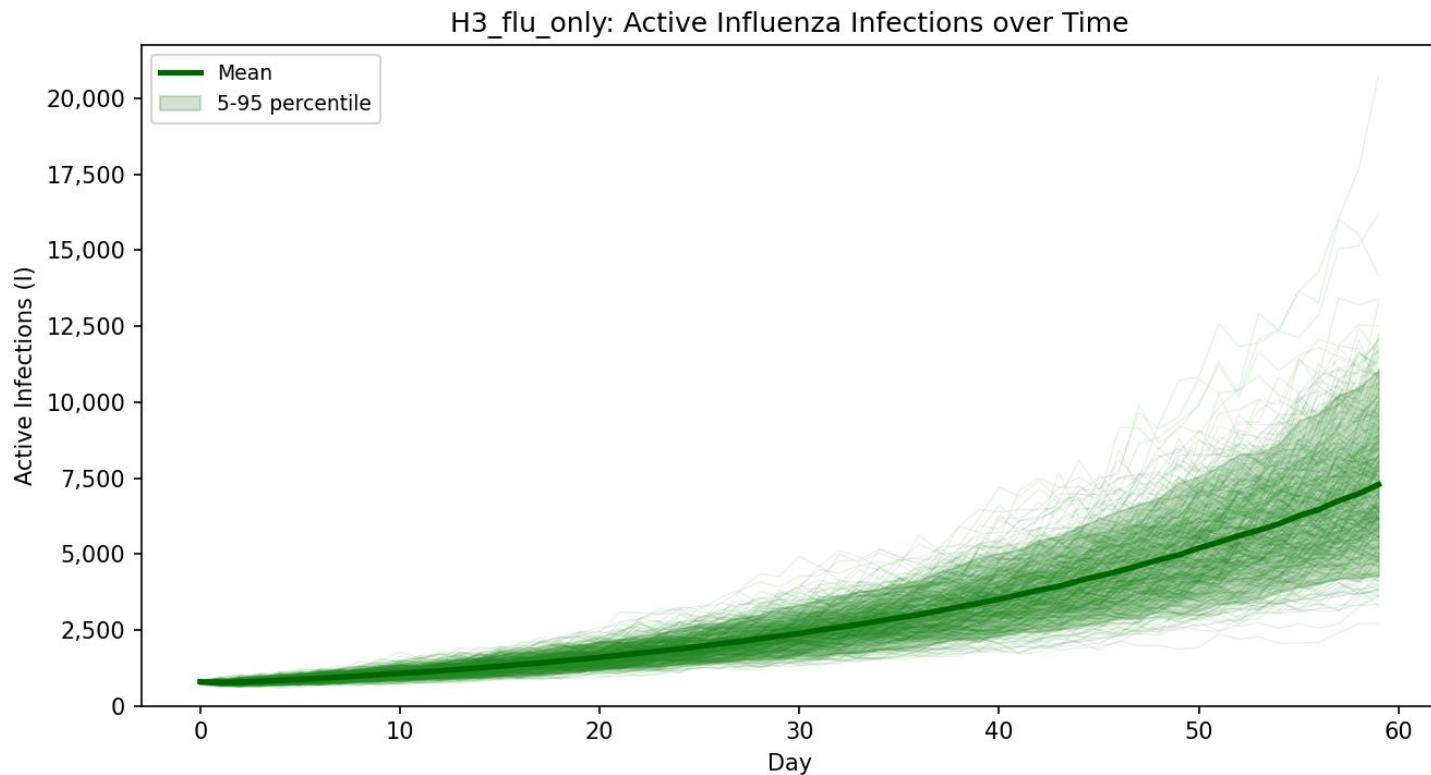
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COVID + Influenza -- Lockdown Impact (3 Variants x Lockdown Levels)



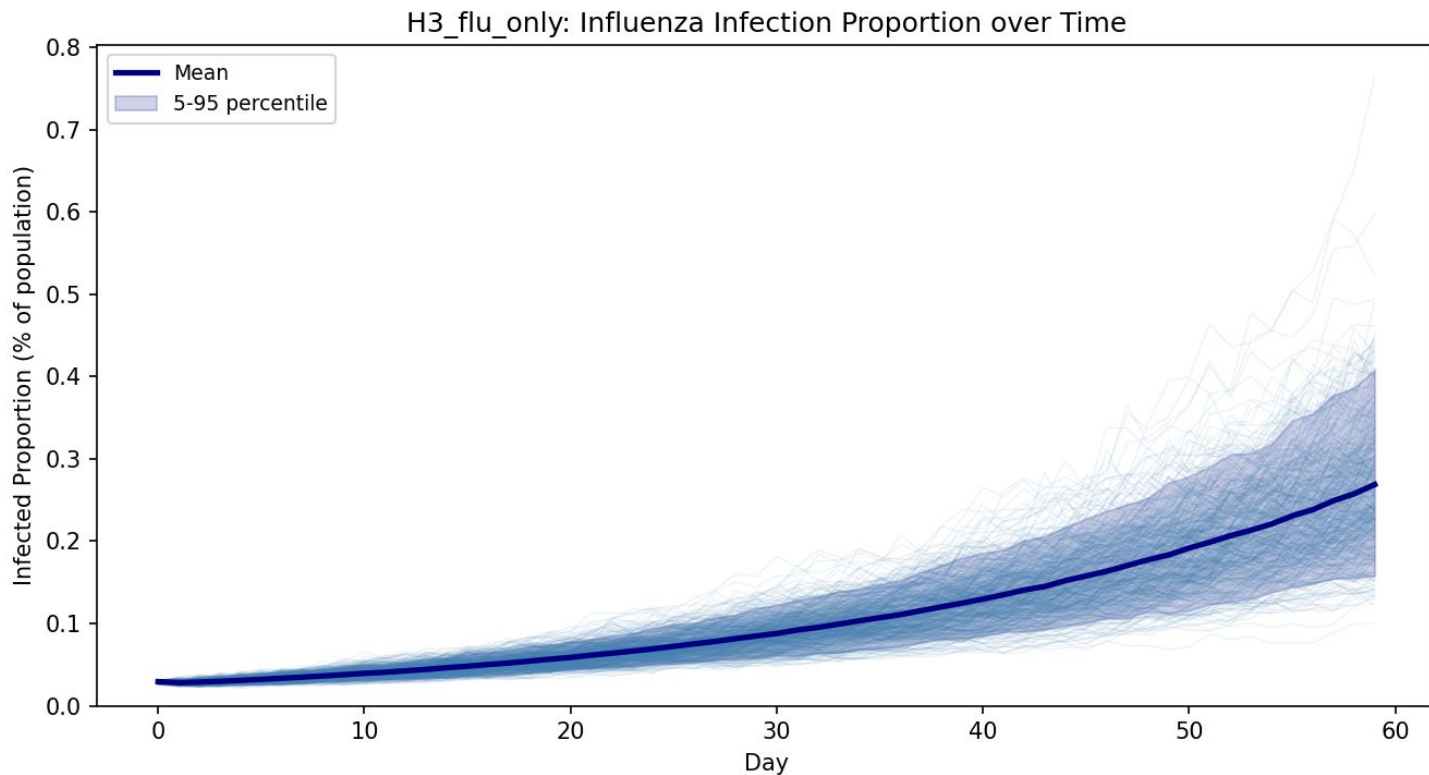
H3 Bonus: The Mystery of the Disappearing Flu Season

Peek (2021), Scientific American: influenza nearly vanished in 2020–21 with no flu-specific intervention



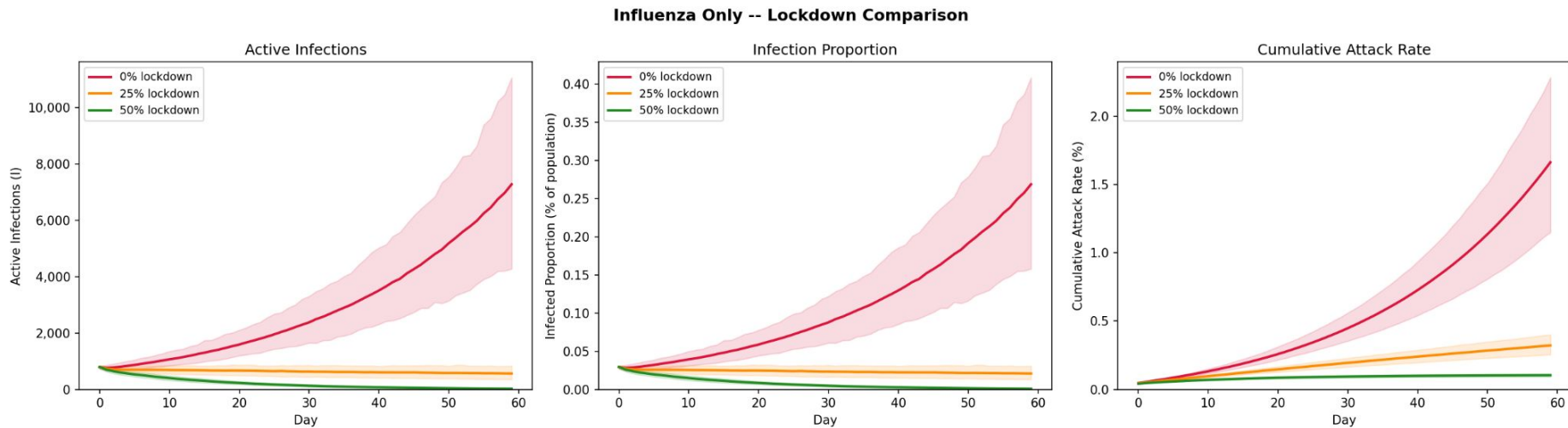
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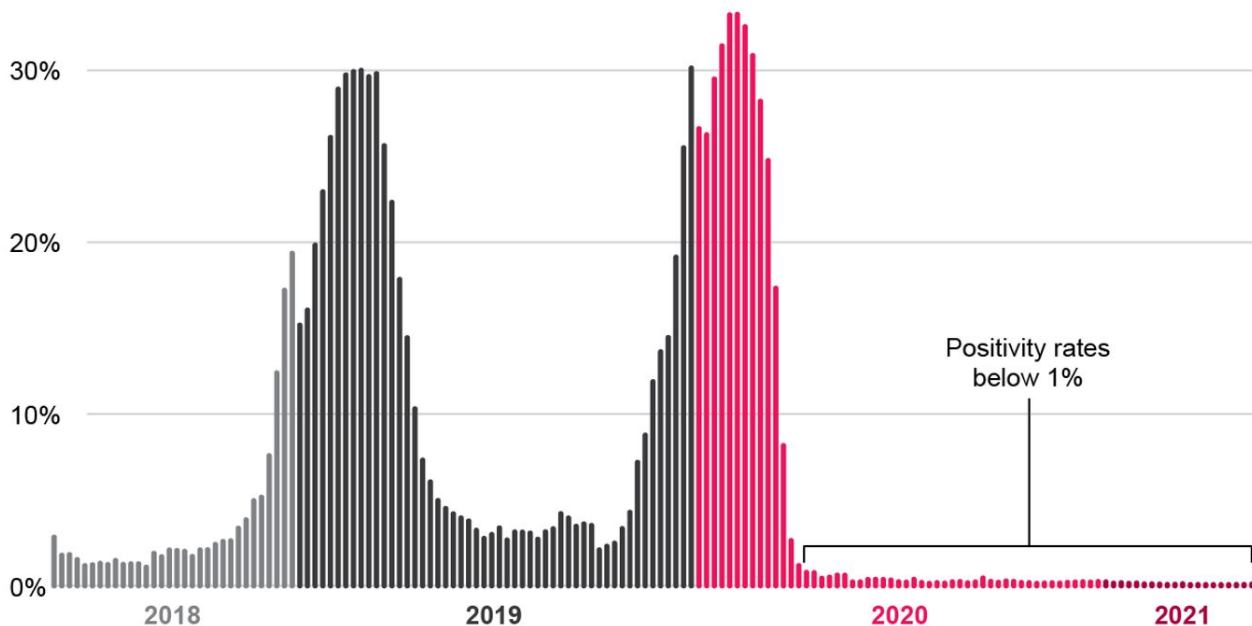


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U.S. Flu Positivity Rates by Week

Since March 2020 fewer people have been tested for influenza, but that is not the reason for fewer recorded cases. The percentage of samples that have tested positive (the positivity rate) has also plummeted.



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Discussion



Key Findings

1 "Elimination" is too strong a claim

Doubling beds cuts overflow probability by $\sim 14\times$ ($83.8\% \rightarrow 6\%$), but doesn't reach zero. The original conclusion was only possible because the bug produced near-zero transmission.

2 Delta has a sharp lockdown threshold

0% lockdown $\rightarrow$ 79.6% overflow. 25% lockdown $\rightarrow$ 0% overflow. There is a critical compliance threshold near 25% where Delta dynamics flip from catastrophic to controlled.

3 Omicron is uncontrollable in this model

Even a 50% lockdown can't prevent overflow at 33,000 beds. It buys time (~ 15 extra days) but can't prevent it. Higher transmissibility overwhelms both bed expansion and behavioural intervention alone.

4 Flu matters – but only at the margin

Flu adds ~ 10 pp to Delta overflow at no lockdown, but becomes irrelevant once the system has headroom (25%+ lockdown) or is already saturated (Omicron). Flu's clinical relevance is regime-dependent.

Limitations & What's Next

Limitations

- ✗ Mass-action mixing — lockdown assumes uniform compliance; real lockdowns have essential workers, households, super-spreader settings
- ✗ No vaccination, waning immunity, or reinfection — the R compartment is absorbing (no second waves)
- ✗ Single-region scope — Chicago parameters don't generalise to other age structures or hospital densities
- ✗ No mass-casualty layer — trauma surge on top of infectious surge was not implemented

Future Work

- Poisson-distributed trauma surge layered on SEIR. Test whether a mass-casualty event tips a marginal system over the edge
- Add vaccination + waning immunity to capture second-wave dynamics
- Finer lockdown sweep around the Delta threshold (e.g., 10%–30%) to precisely localise the critical compliance rate

Thank you!